

CLOUD COMPUTING



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WHAT IS CLOUD COMPUTING??

Cloud computing is the on-demand delivery of IT resources—including computing power, storage, and databases—over the internet with pay-as-you-go pricing. Instead of owning physical data centers, users rent access to services from providers like AWS, Microsoft Azure, or Google Cloud. It is an on-demand, virtualized, and scalable model that replaces upfront capital expenditure with operating expenses.



Characteristics of Cloud Computing

1. **On-Demand Self-Service:** Users can provision computing resources like servers and storage automatically, without requiring human intervention from the service provider.
2. **Broad Network Access:** Capabilities are available over the network and can be accessed through standard mechanisms by any device, such as laptops, tablets, and mobile phones.
3. **Resource Pooling:** The provider's computing resources are pooled to serve multiple customers using a multi-tenant model. Resources are dynamically assigned and reassigned according to demand.
4. **Rapid Elasticity (Scalability):** Resources can be scaled up or down quickly and, in some cases, automatically, to meet demand. This ensures you have enough power during traffic spikes and aren't paying for idle resources during quiet periods.
5. **Measured Service (Pay-as-you-go Model):** Cloud systems automatically control and optimize resource use. Usage is monitored, controlled, and reported, providing transparency for both the provider and the consumer. You only pay for what you use.

MICROSOFT AZURE CLOUD

Microsoft Azure is a public and private cloud platform that enables developers and IT professionals to build, deploy, and manage applications. It relies on virtualization, where a hypervisor abstracts hardware to run multiple virtual machines with different operating systems.

- Uses hypervisors to run multiple virtual machines on a single physical server.
- Supports operating systems like Windows and Linux within virtual environments.
- Operates massive data centers with interconnected servers to ensure scalability and performance.

The 3 Service Categories Provided by Microsoft Azure

Microsoft Azure is a cloud computing platform that offers the following three categories of services:

- Infrastructure as a service (IaaS)
- Platform as a service (PaaS)
- Software as a service (SaaS)

Infrastructure as a service (IaaS)

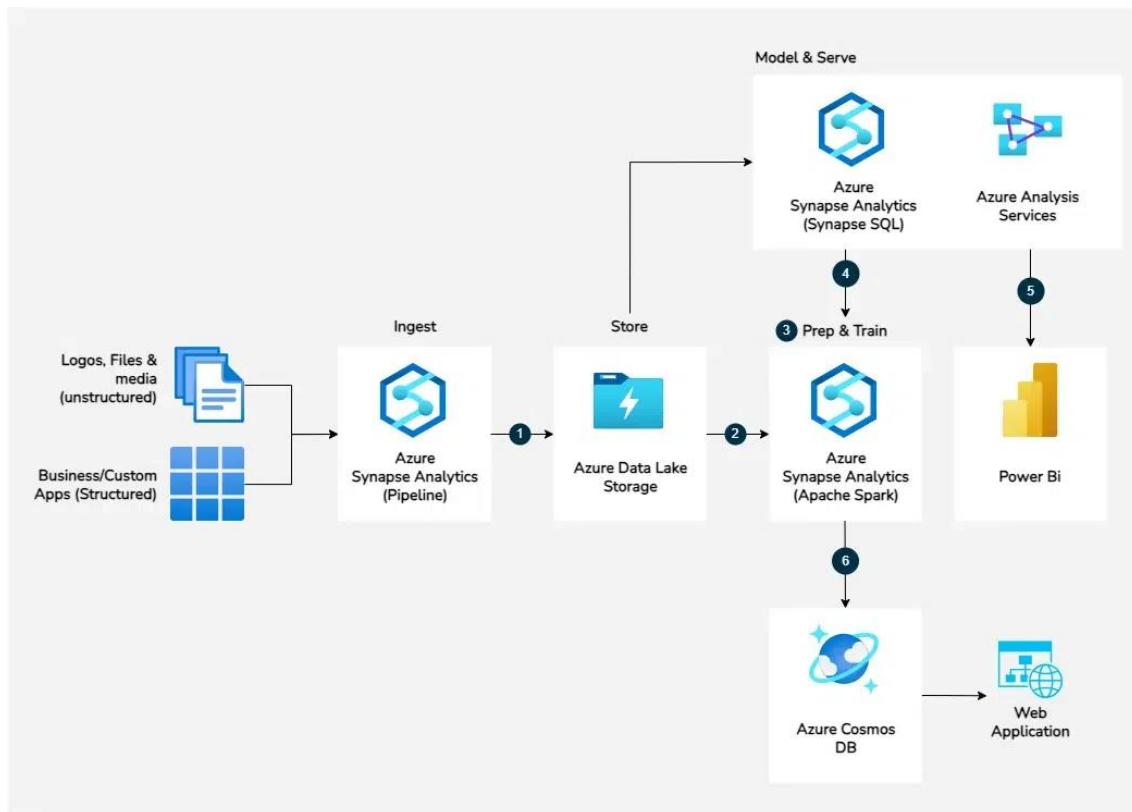
Virtual machines, storage, and networking will come under the category of infrastructure as a service but the users have to do manually the build and deploy of the applications. Azure will support a wide range of operating systems because of its Hyper-hypervisor.

Platform as a service (PaaS)

Azure app service, Azure functions, and logic apps are some services that are offered by Azure under the platform as a service. This service will provide autoscaling and load balancing and also there will be a pre-configured environment for the application.

Software as a service (SaaS)

Office 365, Dynamics 365, and Azure Active Directory are some of the services provided by Microsoft Azure under Software as a Service (SaaS) the complete application will be managed by the Microsoft azure including deploying, scaling and load balancing.



Following are some of the services Microsoft Azure offers:

- **Compute:** Includes Virtual Machines, Virtual Machine Scale Sets, Functions for serverless computing, Batch for containerized batch workloads, Service Fabric for microservices and container orchestration, and Cloud Services for building cloud-based apps and APIs.
- **Networking:** With Azure, you can use a variety of networking tools, like the Virtual Network, which can connect to on-premise data centers; Load Balancer; Application Gateway; VPN Gateway; Azure DNS for domain hosting, Content Delivery Network, Traffic Manager, ExpressRoute dedicated private network fiber connections; and Network Watcher monitoring and diagnostics
- **Storage:** Includes Blob, Queue, File, and Disk Storage, as well as a Data Lake Store, Backup, and Site Recovery, among others.
- **Web + Mobile:** Creating Web + Mobile applications is very easy as it includes several services for building and deploying applications.
- **Containers:** Azure has a property that includes Container Service, which supports [Kubernetes](#), DC/OS or [Docker Swarm](#), and Container Registry, as well as tools for microservices.
- **Databases:** Azure also included several SQL-based databases and related tools.
- **Data + Analytics:** Azure has some big data tools like HDInsight for [Hadoop](#) Spark, R Server, HBase, and Storm clusters

The 7 Core Advantages of Cloud Computing

1. On-Demand Scalability

Maintaining traditional IT infrastructure during economic downturns, rapid expansions, or supply chain disruptions is incredibly difficult. The cloud provides the ability to scale horizontally or vertically at your own speed. Whether you need to aggressively provision resources for a new launch or downscale during a quiet season, you only pay for what you use.

2. Enhanced Security

Small businesses are heavily targeted by cyberattacks, yet many lack the resources to maintain an in-house cybersecurity team. Migrating to the cloud provides immediate access to industry-standard data encryption, enterprise firewalls, and 24/7 automated network monitoring that would otherwise be cost-prohibitive.

3. Access to Modern Technology

The cloud is much more than storage; it is an innovation engine. Businesses can leverage cloud environments to access cutting-edge technologies without buying physical servers. This includes tapping into robust computational power for Artificial Intelligence, Machine Learning models, Big Data Analytics, and containerization.

4. Cost-Effectiveness

Based on the 'pay-as-you-go' utility model, the cloud eliminates the CapEx (Capital Expenditure) of buying servers, hard drives, and networking gear. Instead, it shifts IT spending to a manageable OpEx (Operational Expenditure) model, removing the need to house, cool, and maintain physical data centers.

5. Enterprise Mobility

The cloud untethers the workforce from the physical office. Employees can securely compute heavy tasks, access critical workflows, and manage data from anywhere in the world, fostering a better work-life balance and enabling remote work models.

6. Seamless Collaboration

Cloud services eradicate version-control nightmares. By keeping data in a centralized, cloud-hosted environment, multiple departments, clients, and partners can collaborate simultaneously on the same document or application. This creates a "single source of truth," reducing errors and duplication.

7. Advanced Predictive Analytics

Cloud computing democratizes big data. Businesses can rent massive processing capacities on-demand to run complex predictive analytics. By analyzing historical data, companies can predict market trends, automate low-value processes, and optimize their business strategies based on deep customer insights.

Use Cases of Cloud-Based Services

Cloud-based services are widely used across industries to support modern applications, data management, and digital operations by providing reliable and scalable computing resources over the internet.

- **Web Hosting:** Hosting websites and web applications with high availability and scalability.
- **Application Development:** Building, testing, and deploying applications using cloud platforms and tools.
- **Data Storage and Backup:** Storing large volumes of data securely and creating backups for data recovery.
- **Big Data and Analytics:** Processing and analyzing large datasets efficiently.
- **IoT and AI Workloads:** Supporting Internet of Things devices and artificial intelligence applications.
- **Disaster Recovery:** Restoring data and applications quickly in case of system failures or disasters.
- **Collaboration Tools:** Enabling teams to work together using cloud-based email, file sharing, and communication tools.